

MT WASHINGTON REGIONAL, NH, USA

WMO: 726164

Lat: 44.368N

Lon: 71.545W

Elev: 327

StdP: 97.45

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54728

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.5	(c) -23.7	(d) -31.0	(e) 0.2	(f) -26.5	(g) -27.5	(h) 0.3	(i) -22.9	(j) 9.0	(k) -3.3	(l) 8.1	(m) -5.7	(n) 0.3	(o) 210	(p) 0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	29.9	21.2	28.1	20.1	27.1	19.4	22.5	27.7	21.5	26.4	20.6	25.0	3.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.3	25.0	19.9	15.2	24.1	18.8	14.2	23.3	67.5	27.7	63.9	26.2	60.6	25.1	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.9	(b) 6.9	(c) 5.7	(e) -32.2	(f) 32.4	(g) 4.1	(h) 1.4	(i) -35.1	(j) 33.4	(k) -37.5	(l) 34.2	(m) -39.8	(n) 35.0	(o) -42.8	(p) 36.0		
(a) 7.9	(b) 6.9	(c) 5.7	(e) -31.9	(f) 24.1	(g) 3.5	(h) 0.9	(i) -34.4	(j) 24.8	(k) -36.5	(l) 25.3	(m) -38.5	(n) 25.9	(o) -41.0	(p) 26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-9.5	-8.1	-2.7	4.9	11.6	16.1	18.8	17.7	13.9	7.4	1.2	-5.5		
	DBStd	11.11	7.42	6.83	6.28	4.66	4.27	3.67	3.03	3.16	4.16	4.60	5.24	6.33		
	HDD10.0	2567	603	508	396	164	31	1	0	0	11	108	265	480		
	HDD18.3	4773	862	741	653	402	214	87	31	50	144	339	513	738		
	CDD10.0	947	0	0	1	13	80	184	273	239	126	28	2	0		
	CDD18.3	110	0	0	0	0	4	19	46	31	9	0	0	0		
	CDH23.3	1530	0	0	1	17	108	298	564	386	149	6	0	0		
	CDH26.7	357	0	0	0	2	21	79	141	81	33	0	0	0		
(14) Wind	WSAvg	1.7	1.8	2.0	2.3	2.4	2.0	1.7	1.4	1.2	1.2	1.7	1.7	1.7		
(15) Precipitation	PrecAvg	1063	65	56	63	80	96	113	109	115	90	100	84	85		
	PrecMax	1337	123	120	98	167	153	216	196	207	197	249	147	147		
	PrecMin	764	18	19	27	23	32	34	50	43	34	19	19	39		
	PrecStd	135	24	24	20	36	36	49	34	46	34	58	27	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	12.2	17.9	25.9	29.1	31.5	32.0	31.2	30.3	24.0	18.6	12.3		
		MCWB	9.6	7.9	11.9	14.9	18.8	21.5	23.0	22.0	21.2	17.0	13.5	10.0		
	2%	DB	5.8	7.4	12.7	22.0	26.3	28.8	29.8	28.7	27.1	21.4	15.8	8.2		
		MCWB	3.6	3.8	7.1	12.9	16.2	20.4	21.5	20.8	19.8	15.5	11.5	6.4		
	5%	DB	2.7	3.9	9.0	17.9	23.8	27.0	28.0	27.3	24.8	18.6	12.4	5.4		
		MCWB	1.2	1.5	4.8	10.2	15.2	18.9	20.3	20.0	18.6	13.8	9.1	3.8		
	10%	DB	1.1	2.0	6.3	14.2	21.3	24.5	26.9	25.8	22.5	16.1	9.4	2.9		
		MCWB	-0.1	0.1	2.9	8.3	14.0	17.5	19.6	19.2	17.2	12.3	6.8	1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.1	12.1	16.8	20.1	22.9	24.2	23.3	22.1	18.5	14.8	10.3		
		MCDB	11.2	12.0	17.1	23.0	26.7	28.8	29.5	28.3	27.7	21.6	17.8	12.1		
	2%	WB	3.7	4.2	8.1	14.1	18.4	21.7	22.7	21.9	20.7	16.4	12.3	6.7		
		MCDB	5.1	6.6	11.2	20.4	23.9	27.1	27.7	27.0	25.6	20.0	15.1	8.1		
	5%	WB	1.4	1.8	5.2	11.4	16.7	20.3	21.6	21.0	19.5	14.6	9.7	3.8		
		MCDB	2.5	3.5	8.5	16.5	21.7	24.9	26.4	25.2	23.6	18.1	12.1	5.2		
	10%	WB	-0.2	0.3	3.1	9.0	15.2	18.8	20.6	20.1	18.0	12.6	7.2	1.7		
		MCDB	1.1	1.8	5.8	14.0	19.7	22.8	24.9	24.2	21.3	15.6	9.1	2.8		
(35) Mean Daily Temperature Range	MDBR		11.9	13.3	12.5	12.9	14.2	13.7	13.8	13.8	14.4	12.0	10.3	10.1		
	5% DB	MCDBR	11.9	14.1	16.6	20.6	19.6	17.3	15.8	15.7	17.1	17.0	15.2	11.8		
		MCWBR	10.5	11.3	11.3	11.9	10.5	9.4	8.3	8.7	10.3	11.4	11.4	10.5		
	5% WB	MCDBR	11.8	12.9	14.4	17.6	15.6	14.2	13.6	13.3	14.6	14.8	13.7	11.5		
		MCWBR	10.6	10.5	10.6	11.3	8.9	8.3	7.4	7.7	9.1	10.6	11.1	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.276	0.310	0.344	0.366	0.412	0.420	0.394	0.358	0.322	0.303	0.269		
	taud		2.421	2.336	2.358	2.403	2.402	2.266	2.254	2.321	2.416	2.526	2.514	2.459		
	Ebn at Noon		885	920	928	919	904	859	846	858	866	860	819	830		
	Edh at Noon		72	94	106	111	115	132	132	120	100	79	67	63		
(44) All-Sky Solar Radiation	RadAvg		1.47	2.34	3.50	4.32	4.91	5.36	5.45	4.94	4.03	2.42	1.53	1.12		
	RadStd		0.16	0.25	0.23	0.52	0.60	0.47	0.44	0.28	0.28	0.25	0.19	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.66	N/A	N/A	N/A	N/A	N/A	N/A	-245	+141	+32
(47) Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	N/A

Nomenclature: See separate page